SN74AHCT1G08 Single 2-Input Positive-AND Gate

1 Features

- Operating range: 4.5V to 5.5V
- Maximum t_{pd} of 7.1ns at 5V
- Low power consumption: maximum I_{CC} of 10µA
- ±8mA output drive at 5V
- Inputs are TTL-voltage compatible
- Latch-up performance exceeds 250 mA per JESD 17

2 Applications

- TV, Set-Top Box, and Audio
- Wireless Infrastructure
- Factory Automation and Control
- PC and Notebooks
- Building Automation
- Grid Infrastructure
- Medical, Healthcare, and Fitness
- Printers
- Test and Measurement
- EPOS (Electronic Point of Sale)
- Telecom Infrastructure
- Projectors

3 Description

The SN74AHCT1G08 device is a single 2-input positive-AND gate. The device performs the Boolean function $Y = A \times B$ or $Y = \overline{A + B}$ in positive logic. Low I_{CC} current allows this device to be used in power-sensitive or battery-powered applications.

Package Information

PART NUMBER	PACKAGE(1)	PACKAGE SIZE(2)	BODY SIZE(3)
SN74AHCT1G08	DBV (SOT-23, 5)	2.9mm × 2.8mm	2.9mm × 1.6mm
	DCK (SC70, 5)	2.00mm × 1.25mm	2.00mm × 1.25mm
	DRL (SOT, 5)	1.60mm x 1.6mm	1.60mm x 1.2mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.
(3) The body size (length × width) is a nominal value and does not include pins.



Logic Diagram

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Pin Configuration and Functions

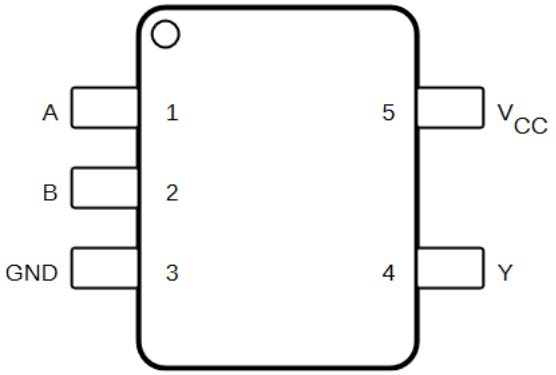


Figure 4-1. DBV, DCK, and DRL Packages 5-Pin SOT-23, SC70, and SOT (Top View)

PIN		TYPE[1]	DESCRIPTION
NAME	NO.		
A	1	I	Input A
B	2	I	Input B
GND	3	—	Ground Pin
V_{CC}	5	—	Supply Pin
Y	4	O	Output

(1) Signal Types: I = Input, O = Output, I/O = Input or Output

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5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted).(1)

			MIN	MAX	UNIT
V_{CC}	Supply voltage		–0.5	7	V
V_I	Input voltage(2)		–0.5	7	V
V_O	Output voltage(2)		–0.5	V_{CC} + 0.5	V
I_{IK}	Input clamp current	V_I < 0		–20	mA
I_{OK}	Output clamp current	V_O < 0 or V_O > V_{CC}		±20	mA
I_O	Continuous output current	V_O = 0 to V_{CC}		±25	mA
	Continuous current through V_{CC} or GND			±50	mA
T_J	Maximum junction temperature			150	°C
T_{stg}	Storage temperature		–65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Section 5.3*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

5.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)	±2500	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101(2)	±1000	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)(1)

		MIN	MAX	UNIT
V_{CC}	Supply voltage	4.5	5.5	V
V_{IH}	High-level input voltage	2		V
V_{IL}	Low-level input voltage		0.8	V
V_I	Input voltage	0	5.5	V
V_O	Output voltage	0	V_{CC}	V
I_{OH}	High-level output current		–8	mA
I_{OL}	Low-level output current		8	mA
Δt/Δv	Input transition rise and fall rate		20	ns/V
T_A	Operating free-air temperature	–40	125	°C

(1) All unused inputs of the device must be held at V_{CC} or GND to ensure proper device operation. See TI application report, *Implications of Slow or Floating CMOS Inputs* (SCBA004).

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5.4 Thermal Information

THERMAL METRIC[1]		SN74AHCT1G08			UNIT
		DBV (SOT-23)	DCK (SC70)	DRL (SOT)	
		5 PINS	5 PINS	5 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	278	289.2	242.9	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	180.5	205.8	77.5	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	184.4	176.2	77.5	°C/W
ψ_{JT}	Junction-to-top characterization parameter	115.4	117.6	9.6	°C/W
ψ_{JB}	Junction-to-board characterization parameter	183.4	175.1	77.3	°C/W
$R_{\theta JC(bot)}$	Junction-to-case (bot) thermal resistance	N/A	N/A	N/A	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, SPRA953.

5.5 Electrical Characteristics

over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
V_{OH}	High-level output voltage	I_{OH} = –50 µA, V_{CC} = 4.5 V	T_A = 25°C	4.4	4.5		V
			T_A = –40°C to 125°C	4.4			
		I_{OH} = –8 mA, V_{CC} = 4.5 V	T_A = 25°C	3.94			
			T_A = –40°C to 125°C	3.8			
V_{OL}	Low-level output voltage	I_{OL} = 50 µA, V_{CC} = 4.5 V				0.1	V
		I_{OL} = 8 mA, V_{CC} = 4.5 V	T_A = 25°C			0.36	
			T_A = –40°C to 125°C			0.44	
I_I	Input current	V_I = 5.5 V or GND, V_{CC} = 0 V to 5.5 V	T_A = 25°C			±0.1	µA
			T_A = –40°C to 125°C			±1	
I_{CC}	Supply current	V_I = V_{CC} or GND, I_O = 0, V_{CC} = 5.5 V	T_A = 25°C			1	µA
			T_A = –40°C to 125°C			10	
ΔI_{CC} [1]	Change in supply current	One input at 3.4 V, Other Inputs at V_{CC} or GND, V_{CC} = 5.5 V	T_A = 25°C			1.35	mA
			T_A = –40°C to 125°C			1.5	
C_I	Input capacitance	V_I = V_{CC} or GND, V_{CC} = 5 V			4	10	pF

(1) This is the increase in supply current for each input at one of the specified TTL voltage levels, rather than 0 V or V_{CC}.

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5.6 Switching Characteristics

over recommended operating free-air temperature range, V_{CC} = 5 V ± 0.5 V (unless otherwise noted) (see Load Circuit and Voltage Waveforms)

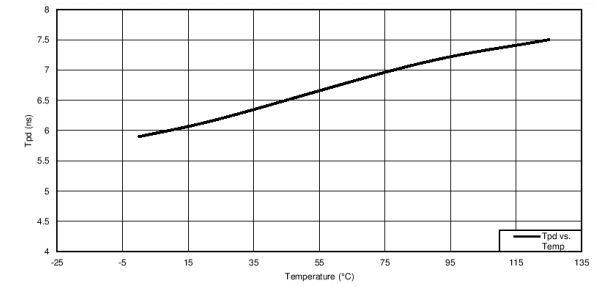
PARAMETER		FROM (INPUT)	TO (OUTPUT)	OUTPUT CAPACITANCE	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{PLH}	Propagation delay, low to high transition	A or B	Y	C_L = 15 pF	T_A = 25°C		5	6.2	ns
					T_A = –40°C to 85°C	1		7.1	
					T_A = –40°C to 125°C	1		7.5	
t_{PHL}	Propagation delay, high to low transition	A or B	Y	C_L = 15 pF	T_A = 25°C		5	6.2	ns
					T_A = –40°C to 85°C	1		7.1	
					T_A = –40°C to 125°C	1		7.5	
t_{PLH}	Propagation delay, low to high transition	A or B	Y	C_L = 50 pF	T_A = 25°C		5.5	7.9	ns
					Propagation delay, high to low transition	1		9	
					T_A = –40°C to 125°C	1		10	
t_{PHL}	Propagation delay, high to low transition	A or B	Y	C_L = 50 pF	T_A = 25°C		5.5	7.9	ns
					T_A = –40°C to 85°C	1		9	
					T_A = –40°C to 125°C	1		10	

5.7 Operating Characteristics

V_{CC} = 5 V, T_A = 25°C

PARAMETER		TEST CONDITIONS	TYP	UNIT
C_{pd}	Power dissipation capacitance	No load, f = 1 MHz	18	pF

5.8 Typical Characteristics

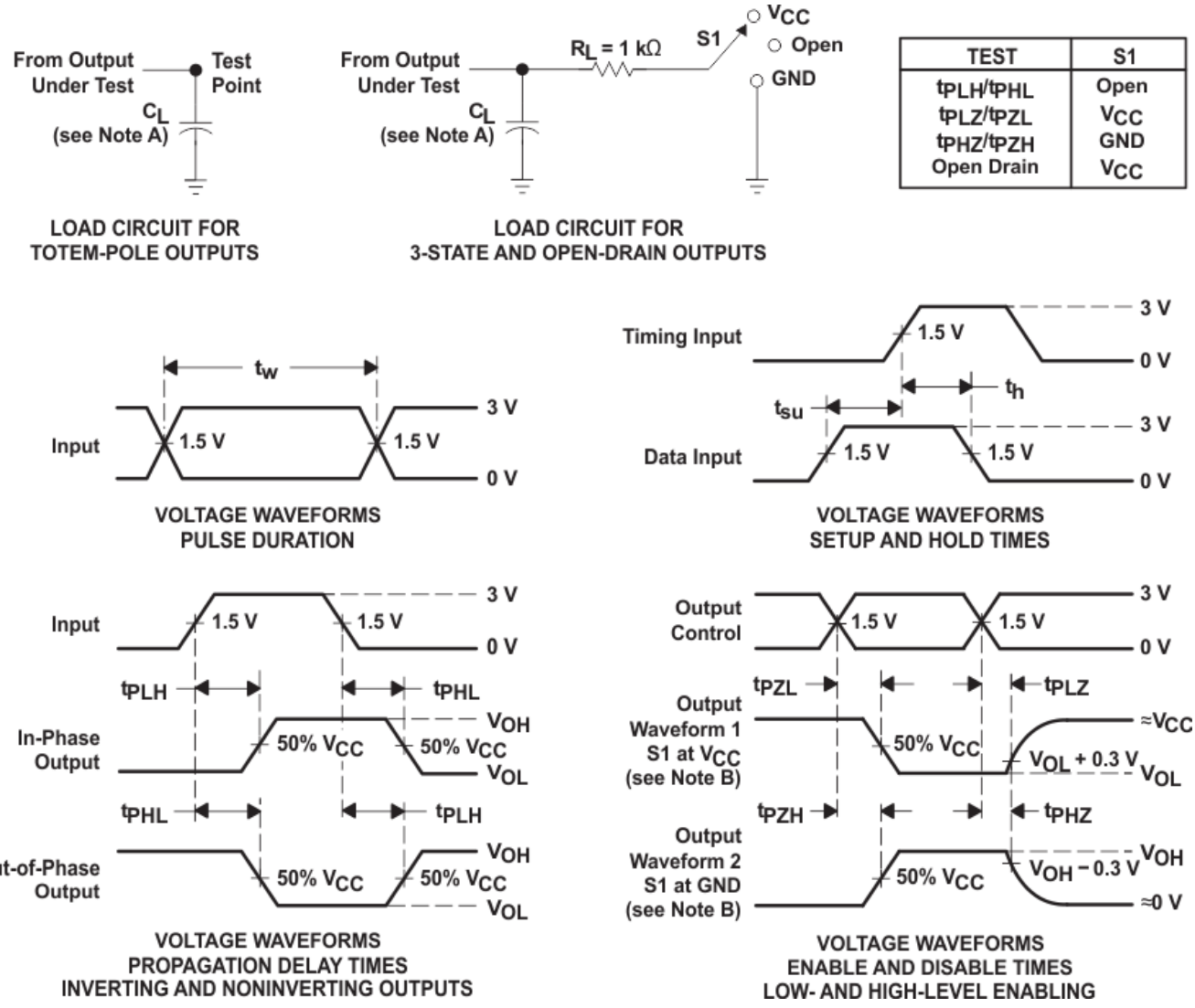


C_L = 15 pF

Figure 5-1. T_{pd} vs Temperature

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6 Parameter Measurement Information



C_L includes probe and jig capacitance.
Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control.
Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
All input pulses are supplied by generators having the following characteristics: PRR ≤ 1 MHz, Z_O = 50 Ω, t_r ≤ 3 ns, t_f ≤ 3 ns.
The outputs are measured one at a time with one input transition per measurement.
All parameters and waveforms are not applicable to all devices.

Figure 6-1. Load Circuit and Voltage Waveforms

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7 Detailed Description

7.1 Overview

The SN74AHCT1G08 device is a single 2-input positive-AND gate. The device performs the Boolean AND function ($Y = A \bullet B$ or $Y = \overline{\overline{A} + \overline{B}}$) in positive logic. Low I_{CC} current allows this device to be used in power-sensitive or battery-powered applications. Robust inputs allow the device to up-translate with a propagation delay of 20 ns.

7.2 Functional Block Diagram



Figure 7-1. Logic Diagram (Positive Logic)

7.3 Feature Description

The V_{CC} for the device is optimized at 5 V.

Up voltage translation from 3.3 V to 5 V is allowed. The inputs accept V_{IH} levels of 2 V.

Output ringing is minimized by slow edge rates.

Inputs are TTL-Voltage compatible.

7.4 Device Functional Modes

Table 7-1 lists the functional modes of the SN74AHCT1G08.

Table 7-1. Function Table

INPUTS(1)		OUTPUT(2)
A	B	Y
H	H	H
L	X	L
X	L	L

(1) H = High Voltage Level, L = Low Voltage Level, X = Don't Care
(2) H = Driving High, L = Driving Low, Z = High Impedance State

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8 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

8.1 Application Information

The SN74AHCT1G08 device is a single AND gate, which is often used for many common functions like power sequencing or an *on* LED indicator. Because the device is configured to output LOW unless all inputs are HIGH, an LED tied to the output of the device will only turn HIGH when all systems connected are sending a HIGH, or *ready* signal.

8.2 Typical Application

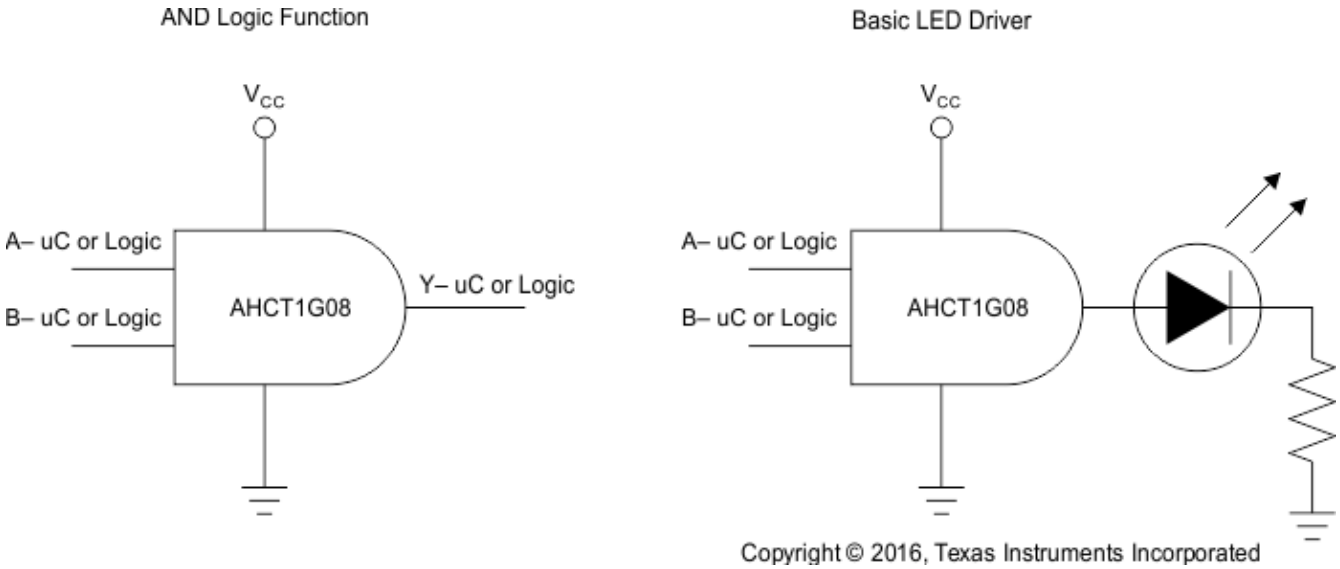


Figure 8-1. Typical Application Diagram

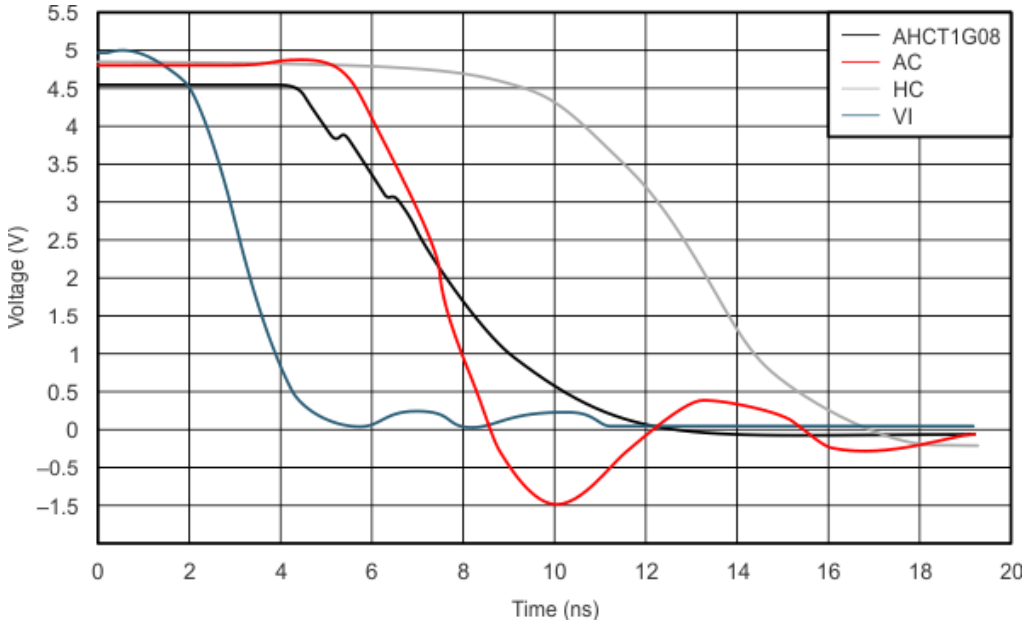
8.2.1 Design Requirements

This device uses CMOS technology and has balanced output drive. Take care to avoid bus contention because it can drive currents that would exceed maximum limits. The high drive also creates fast edges into light loads, so routing and load conditions must be considered to prevent ringing.

8.2.2 Detailed Design Procedure

1. Recommended Input Conditions
   - For rise time and fall time specifications, see Δt/ΔV in *Section 5.3*.
   - For specified high and low levels, see V_{IH} and V_{IL} in *Section 5.3*.
   - Inputs are overvoltage tolerant allowing them to go as high as 5.5 V at any valid V_{CC}.
2. Recommended Output Conditions
   - Load currents must not exceed 25 mA per output and 50 mA total for the part.
   - Outputs must not be pulled above V_{CC}.

8.2.3 Application Curve



V_{CC} = 5 V Load = 50 Ω / 50 pF

Figure 8-2. Typical Switching Characteristics

8.3 Power Supply Recommendations

The power supply can be any voltage between the minimum and maximum supply voltage rating located in *Recommended Operating Conditions*.

Each V_{CC} pin must have a good bypass capacitor to prevent power disturbance. TI recommends a 0.1-µF capacitor for devices with a single supply; and a 0.01-µF or 0.022-µF capacitor for each power pin if there are multiple V_{CC} pins. It is ok to parallel multiple bypass capacitors to reject different frequencies of noise. 0.1-µF and 1-µF capacitors are commonly used in parallel. The bypass capacitor must be installed as close to the power pin as possible for best results.

8.4 Layout

8.4.1 Layout Guidelines

When using multiple bit logic devices inputs must not ever float. In many cases, functions or parts of functions of digital logic devices are unused; for example, when only two inputs of a triple-input AND gate are used or only 3 of the 4 buffer gates are used. Such input pins must not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. Observe the following rules under all circumstances.

- All unused inputs of digital logic devices must be connected to a high or low bias to prevent them from floating.
- The logic level that must be applied to any particular unused input depends on the function of the device. Generally they will be tied to GND or V_{CC}, whichever make more sense or is more convenient.

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8.4.1.1 Layout Example

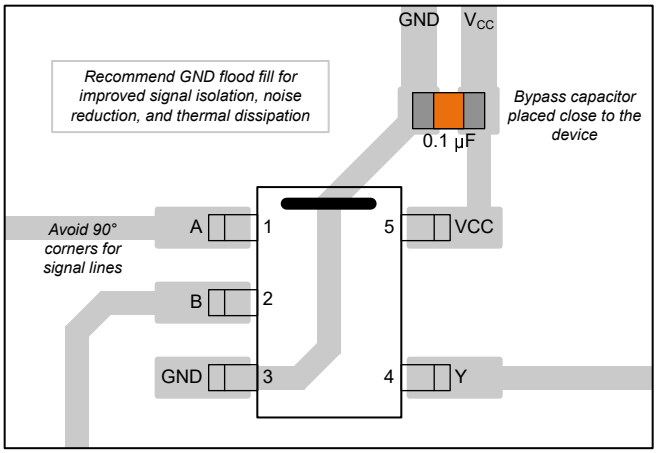


Figure 8-3. Layout Example for the SN74AHCT1G08

9 Device and Documentation Support

9.1 Documentation Support (Analog)

9.1.1 Related Documentation

For related documentation, see the following:

Implications of Slow or Floating CMOS Inputs, SCBA004

9.2 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's Terms of Use.

9.3 Trademarks

TI E2E™ is a trademark of Texas Instruments.
All trademarks are the property of their respective owners.

9.4 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

9.5 Glossary

TI Glossary	This glossary lists and explains terms, acronyms, and definitions.

10 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision R (October 2023) to Revision S (February 2024)	Page
• Updated thermal values for DBV package from RθJA = 226 to 278, RθJC(top) = 165 to 180.5, RθJB = 59.1 to 184.4, ΨJT = 45.5 to 115.4, ΨJB = 58.3 to 183.4, RθJC(bot) = N/A, all values in °C/W	5

Changes from Revision Q (April 2016) to Revision R (October 2023)	Page
• Updated the numbering format for tables, figures, and cross-references throughout the document.	1
• Updated thermal values for DCK package from RθJA = 277.5 to 289.2, RθJC(top) = 92.9 to 205.8, RθJB = 64.2 to 176.2, ΨJT = 1.9 to 117.6, ΨJB = 63.5 to 175.1, RθJC(bot) = N/A, all values in °C/W	5

11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser based versions of this data sheet, refer to the left hand navigation.

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PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
74AHCT1G08DBVRG4	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	B08G
74AHCT1G08DBVRG4.A	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	B08G
74AHCT1G08DBVTG4.A	Active	Production	SOT-23 (DBV) | 5	250 | SMALL T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	B08G
74AHCT1G08DCKRE4	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BE3, BEG, BEL, BES)
74AHCT1G08DCKRE4	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	No	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BE3, BEG, BEL, BES)
74AHCT1G08DCKRE4.A	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BE3, BEG, BEL, BES)
74AHCT1G08DCKRE4.A	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	No	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BE3, BEG, BEL, BES)
74AHCT1G08DCKRG4	Active	Production	SC70 (DCK) | 5	3000 | null	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BE3, BEG, BEL, BES)
74AHCT1G08DCKRG4	Active	Production	SC70 (DCK) | 5	3000 | null	No	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BE3, BEG, BEL, BES)
74AHCT1G08DCKRG4.A	Active	Production	SC70 (DCK) | 5	3000 | null	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BE3, BEG, BEL, BES)
74AHCT1G08DCKRG4.A	Active	Production	SC70 (DCK) | 5	3000 | null	No	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BE3, BEG, BEL, BES)
74AHCT1G08DRLRG4	Active	Production	SOT-5X3 (DRL) | 5	4000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BEB, BES)
SN74AHCT1G08DBV3	Last Time Buy	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	SNBI	Level-1-260C-UNLIM	-40 to 125	B08Y
SN74AHCT1G08DBV3.A	Last Time Buy	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	SNBI	Level-1-260C-UNLIM	-40 to 125	B08Y
SN74AHCT1G08DBVR	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(38WH, 3C8F, B083, B08G, B08J, B08L, B08S)
SN74AHCT1G08DBVR.A	Active	Production	SOT-23 (DBV) | 5	3000 | LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(38WH, 3C8F, B083, B08G, B08J, B08L, B08S)
SN74AHCT1G08DBVT	Obsolete	Production	SOT-23 (DBV) | 5	-	-	Call TI	Call TI	-40 to 125	(B083, B08G, B08J, B08S)

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
SN74AHCT1G08DCK3	Last Time Buy	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	SNBI	Level-1-260C-UNLIM	-40 to 125	BEY
SN74AHCT1G08DCK3.A	Last Time Buy	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	SNBI	Level-1-260C-UNLIM	-40 to 125	BEY
SN74AHCT1G08DCKR	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(1PJ, BE3, BEG, BEJ, BEL, BES)
SN74AHCT1G08DCKR.A	Active	Production	SC70 (DCK) | 5	3000 | LARGE T&R	Yes	SN	Level-1-260C-UNLIM	-40 to 125	(1PJ, BE3, BEG, BEJ, BEL, BES)
SN74AHCT1G08DCKT	Obsolete	Production	SC70 (DCK) | 5	-	-	Call TI	Call TI	-40 to 125	(BE3, BEG, BES)
SN74AHCT1G08DRLR	Active	Production	SOT-5X3 (DRL) | 5	4000 | LARGE T&R	Yes	NIPDAU | NIPDAUAG	Level-1-260C-UNLIM	-40 to 125	(BEB, BES)
SN74AHCT1G08DRLR.A	Active	Production	SOT-5X3 (DRL) | 5	4000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	(BEB, BES)

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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OTHER QUALIFIED VERSIONS OF SN74AHCT1G08 :

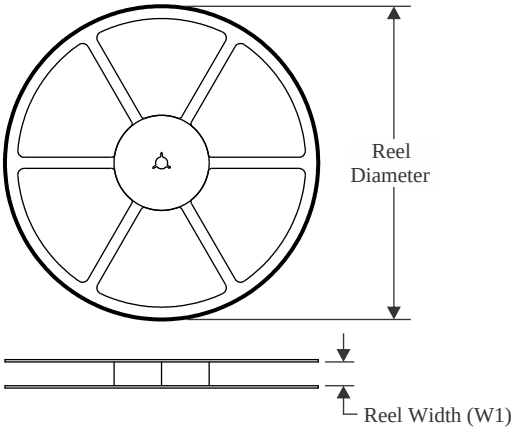
- Automotive : SN74AHCT1G08-Q1

NOTE: Qualified Version Definitions:

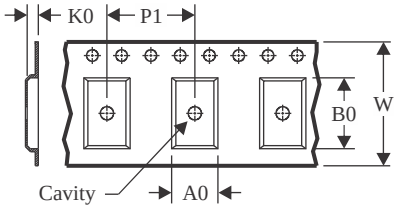
- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects

TAPE AND REEL INFORMATION

REEL DIMENSIONS

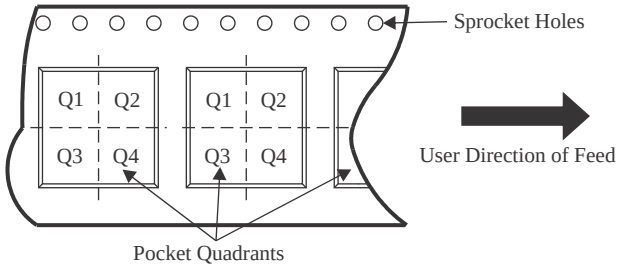


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

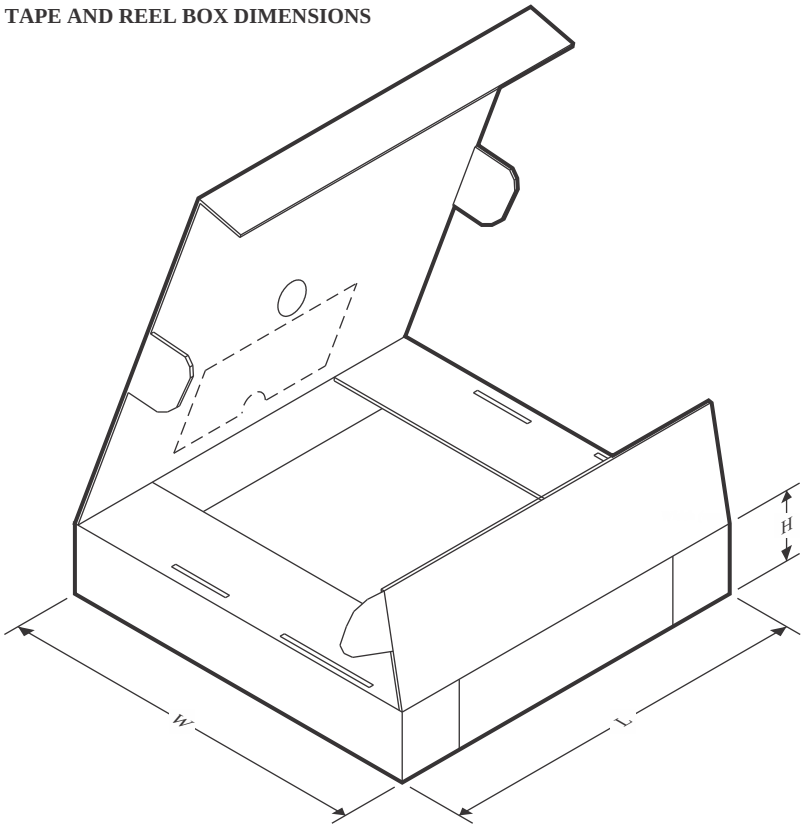
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

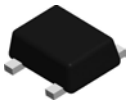
Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
74AHCT1G08DBVRG4	SOT-23	DBV	5	3000	178.0	9.0	3.23	3.17	1.37	4.0	8.0	Q3
74AHCT1G08DCKRE4	SC70	DCK	5	3000	180.0	8.4	2.3	2.55	1.2	4.0	8.0	Q3
74AHCT1G08DCKRE4	SC70	DCK	5	3000	178.0	9.2	2.4	2.4	1.22	4.0	8.0	Q3
74AHCT1G08DCKRE4	SC70	DCK	5	3000	180.0	8.4	2.47	2.3	1.25	4.0	8.0	Q3
74AHCT1G08DCKRE4	SC70	DCK	5	3000	178.0	9.0	2.4	2.5	1.2	4.0	8.0	Q3
SN74AHCT1G08DBVR	SOT-23	DBV	5	3000	178.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74AHCT1G08DBVR	SOT-23	DBV	5	3000	180.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74AHCT1G08DBVR	SOT-23	DBV	5	3000	180.0	8.4	3.2	3.2	1.4	4.0	8.0	Q3
SN74AHCT1G08DCKR	SC70	DCK	5	3000	178.0	9.0	2.4	2.5	1.2	4.0	8.0	Q3
SN74AHCT1G08DCKR	SC70	DCK	5	3000	180.0	8.4	2.3	2.55	1.2	4.0	8.0	Q3
SN74AHCT1G08DRLR	SOT-5X3	DRL	5	4000	180.0	8.4	2.0	1.8	0.75	4.0	8.0	Q3
SN74AHCT1G08DRLR	SOT-5X3	DRL	5	4000	180.0	8.4	1.98	1.78	0.69	4.0	8.0	Q3

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
74AHCT1G08DBVRG4	SOT-23	DBV	5	3000	180.0	180.0	18.0
74AHCT1G08DCKRE4	SC70	DCK	5	3000	205.0	200.0	33.0
74AHCT1G08DCKRE4	SC70	DCK	5	3000	180.0	180.0	18.0
74AHCT1G08DCKRE4	SC70	DCK	5	3000	202.0	201.0	28.0
74AHCT1G08DCKRE4	SC70	DCK	5	3000	180.0	180.0	18.0
SN74AHCT1G08DBVR	SOT-23	DBV	5	3000	208.0	191.0	35.0
SN74AHCT1G08DBVR	SOT-23	DBV	5	3000	210.0	185.0	35.0
SN74AHCT1G08DBVR	SOT-23	DBV	5	3000	210.0	185.0	35.0
SN74AHCT1G08DCKR	SC70	DCK	5	3000	180.0	180.0	18.0
SN74AHCT1G08DCKR	SC70	DCK	5	3000	210.0	185.0	35.0
SN74AHCT1G08DRLR	SOT-5X3	DRL	5	4000	210.0	185.0	35.0
SN74AHCT1G08DRLR	SOT-5X3	DRL	5	4000	202.0	201.0	28.0

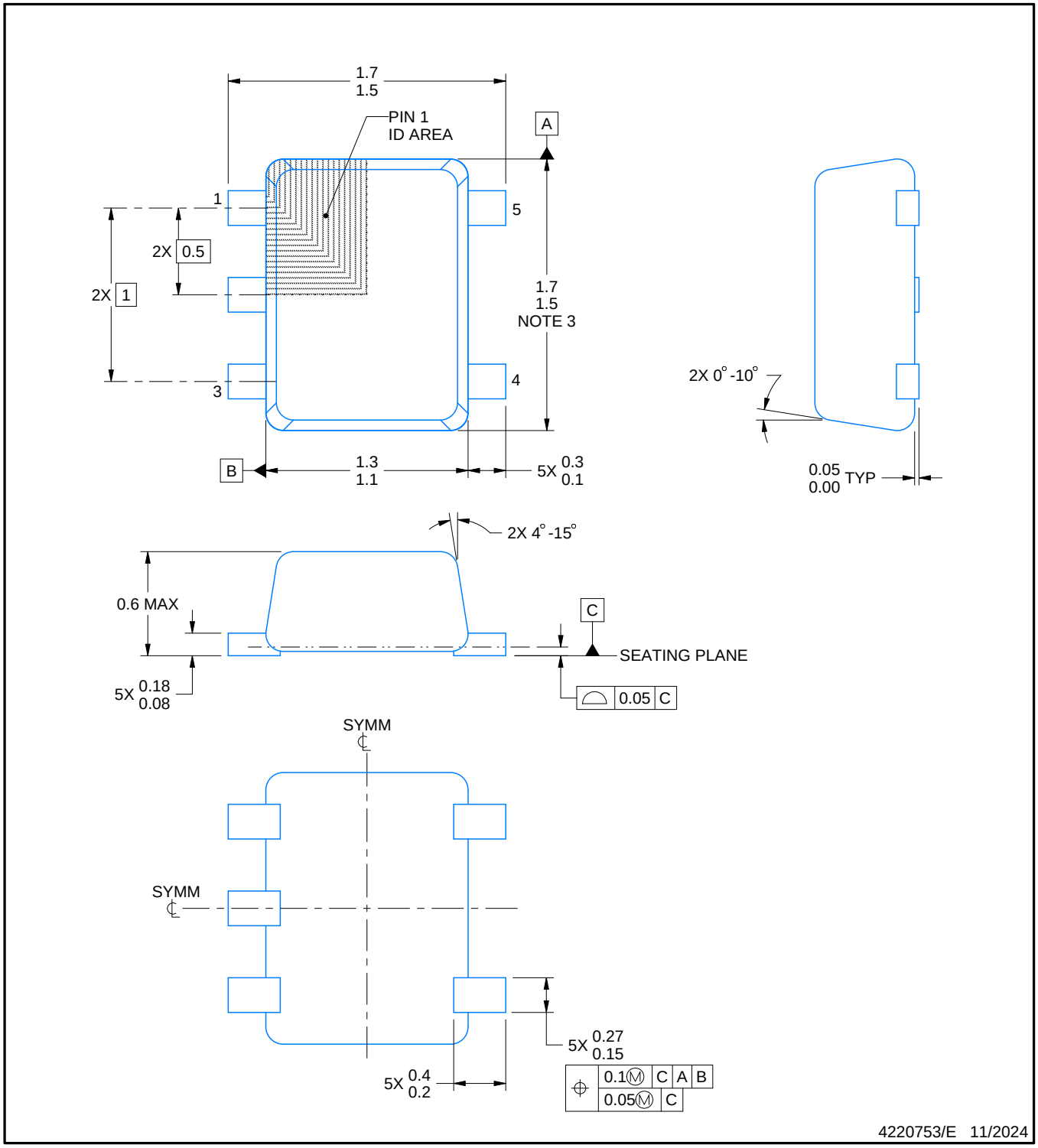


PACKAGE OUTLINE

DRL0005A

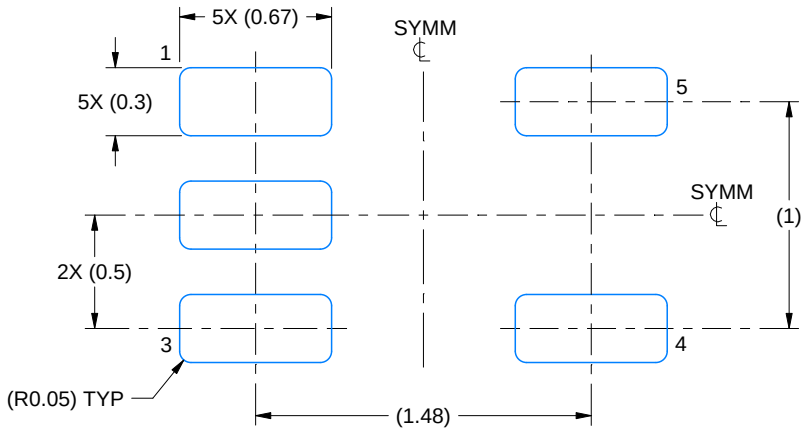
SOT - 0.6 mm max height

PLASTIC SMALL OUTLINE

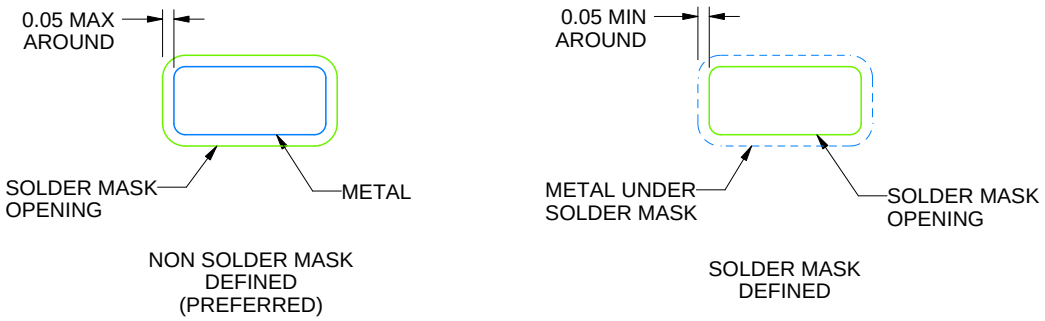


NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-293 Variation UAAD-1



LAND PATTERN EXAMPLE
SCALE:30X

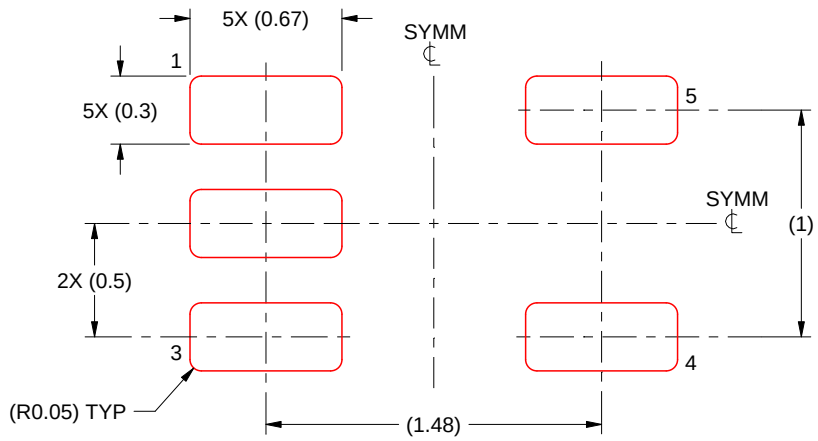


SOLDERMASK DETAILS

4220753/E 11/2024

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.1 mm THICK STENCIL
SCALE:30X

4220753/E 11/2024

NOTES: (continued)

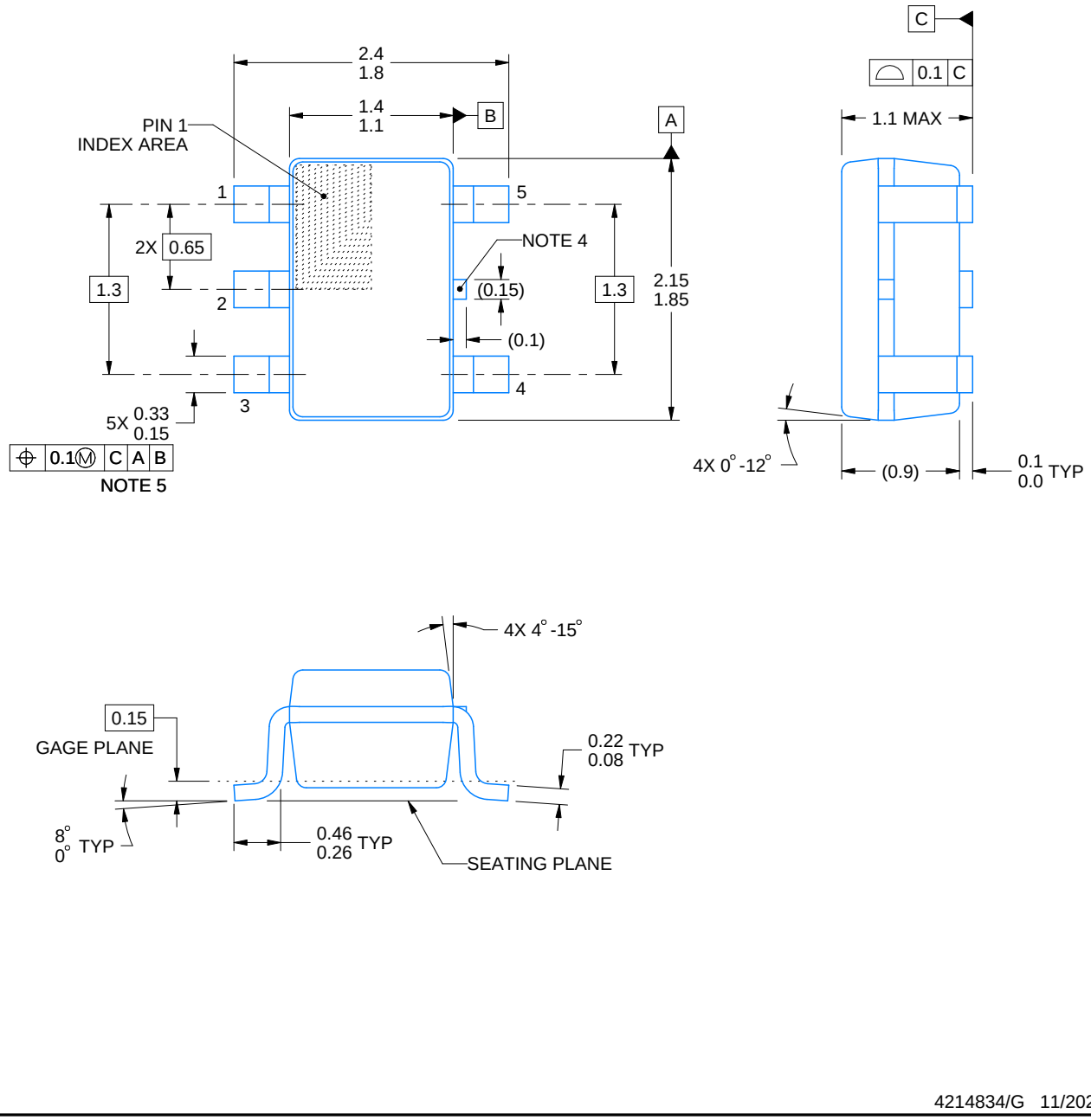
7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

PACKAGE OUTLINE

DCK0005A

SOT - 1.1 max height

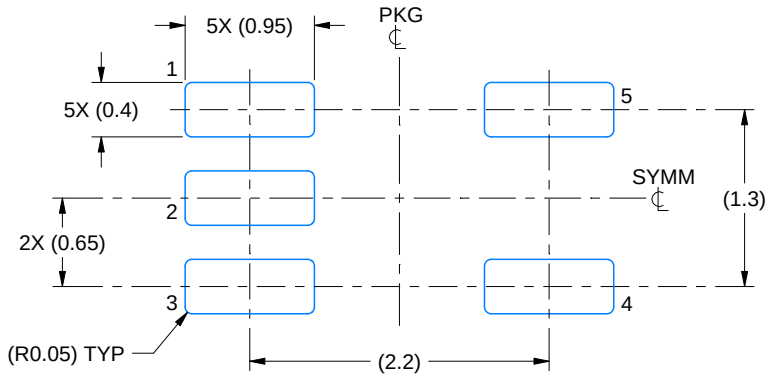
SMALL OUTLINE TRANSISTOR



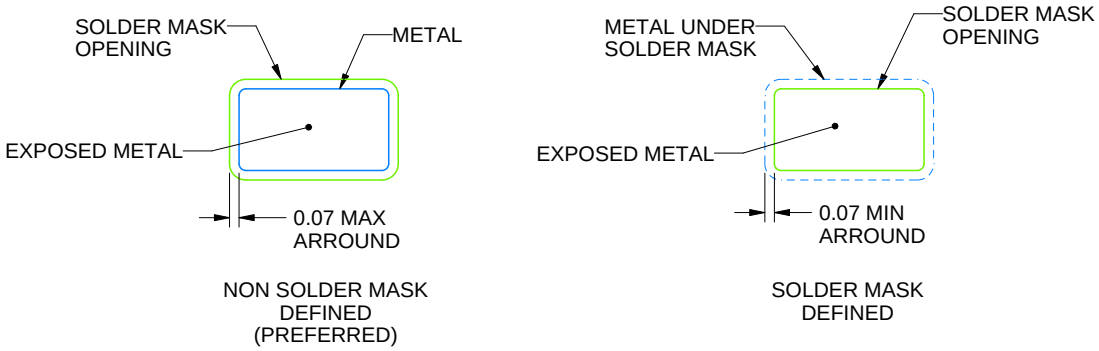
4214834/G 11/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Refernce JEDEC MO-203.
4. Support pin may differ or may not be present.
5. Lead width does not comply with JEDEC.
6. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.25mm per side



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:18X

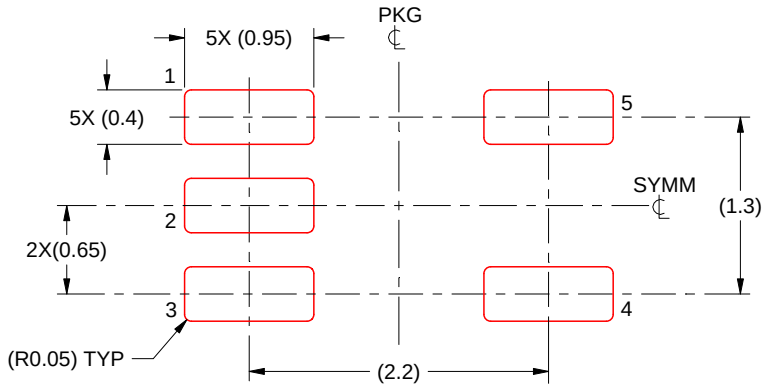


SOLDER MASK DETAILS

4214834/G 11/2024

NOTES: (continued)

7. Publication IPC-7351 may have alternate designs.
8. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 THICK STENCIL
SCALE:18X

4214834/G 11/2024

NOTES: (continued)

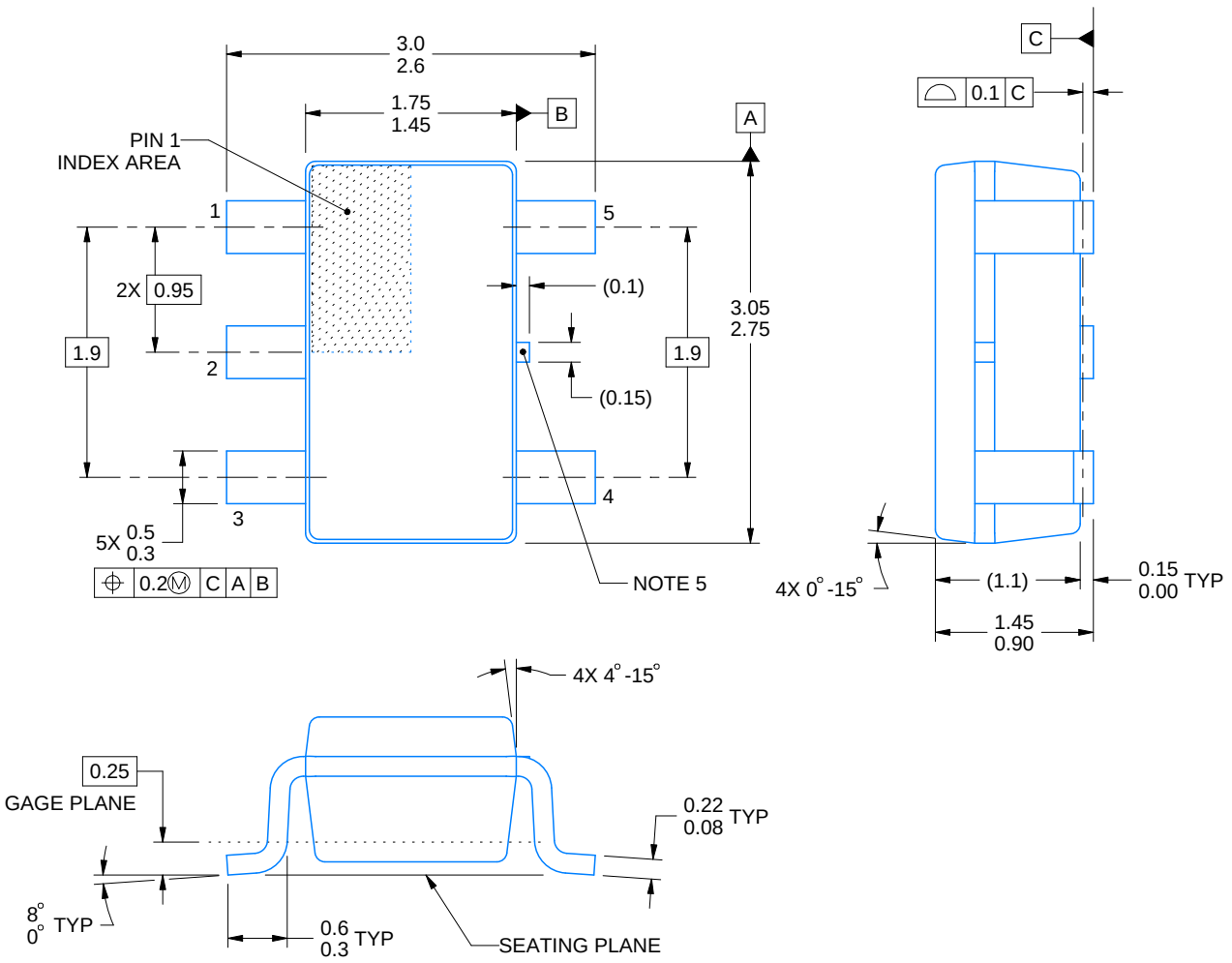
9. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
10. Board assembly site may have different recommendations for stencil design.

PACKAGE OUTLINE

DBV0005A

SOT-23 - 1.45 mm max height

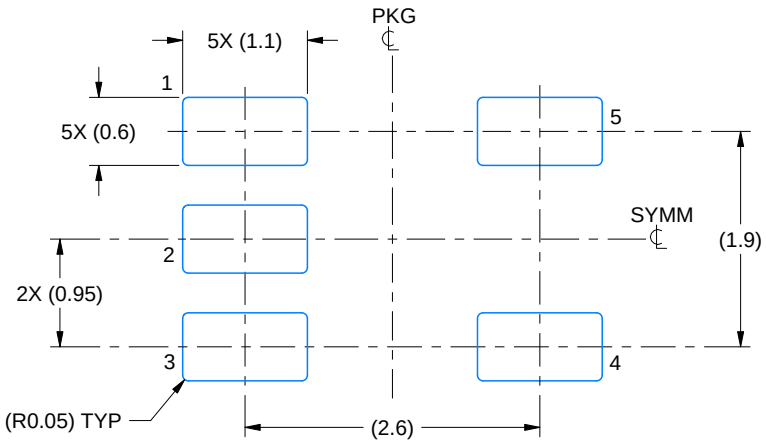
SMALL OUTLINE TRANSISTOR



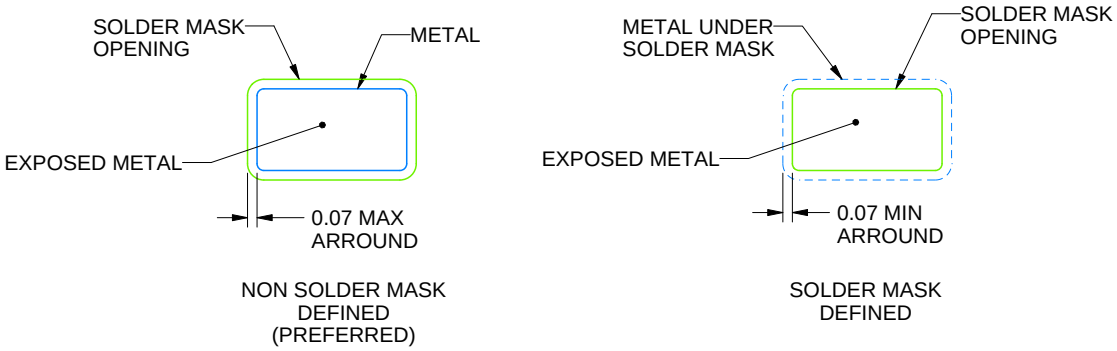
4214839/K 08/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Refernce JEDEC MO-178.
4. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.25 mm per side.
5. Support pin may differ or may not be present.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:15X

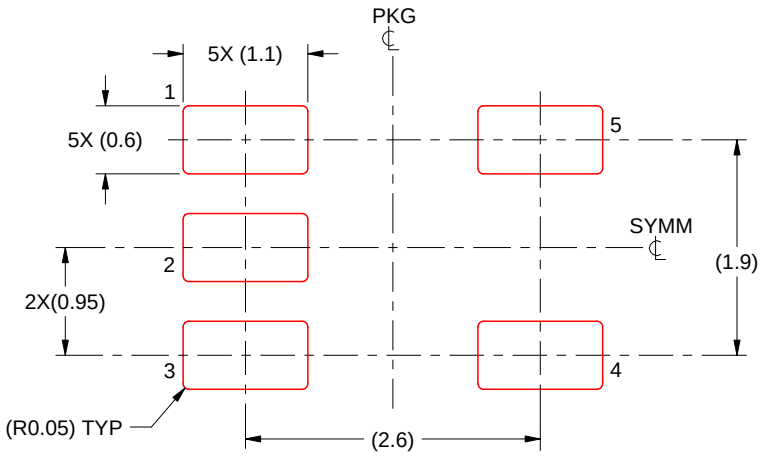


SOLDER MASK DETAILS

4214839/K 08/2024

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:15X

4214839/K 08/2024

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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